

Course title: CST5020 - Convex Optimization
Term: Spring 2026
Credits: 1
Time: 9:50 - 11:25 on Wednesdays
Location: E10-306
Website: <https://chemaoxfz.github.io/cvxopt>

Course instructor: Fangzhou Xiao
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Course description

Introduction to concepts and methods in optimization, especially convex optimization. Focuses on formulating problems in applications into optimization problems. The basic analysis methods for optimization problems needed for this goal will be covered, such as optimality conditions, duality theory, theorems of alternatives, and so on. Mathematical maturity and familiarity with linear algebra is preferred. Prerequisite: Multi-variable Calculus Track A/B, Linear Algebra II Track A/B, Probability and Statistics/Probability and Statistics A/B.

Course schedule

Lecture #	Topic
1	Introduction, Convex sets
2	Convex functions, Convex optimization problems
3	Duality
4	Approximation and fitting
5	Statistical estimation, geometric problems
6	Unconstrained minimization
7	Equality constrained minimization
8	Interior point method

Learning objectives

- To understand the structure of optimization problems so as to formulate them to solve problems that arise in applications
- To have tools of analysis for formulated convex optimization problems
- To understand how convex optimization problems are solved, computationally
- To have experience in solving some convex optimization problems.

Learning resources

- Textbook: Stephen Boyd and Lieven Vandenberghe. *Convex optimization*.
 - o Available at this link: <https://web.stanford.edu/~boyd/cvxbook/>
- The course that we will mostly follow for the basics of convex optimization:
 - o <https://web.stanford.edu/class/ee364a/lectures.html>
- Another course that introduces convex optimization with a more machine learning perspective:
 - o <https://www.stat.cmu.edu/~ryantibs/convexopt/>
- For some background on basics of linear algebra: Stephen Boyd and Lieven Vandenberghe. *Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares*
- Further references on a different presentation on convex optimization: Dimitri P. Bertsekas. *Convex Optimization Theory*, 2009

Assignments

- There are no official assignments of the class. Problem sets and other notes may be handed out to facilitate your learning, but not graded.
- You are encouraged to reinforce your understanding by working out the examples and exercises in the textbook, or apply the methods learned to problems you're interested in.
- A good skill to have for writing documents involving significant math is latex. For intro to latex, overleaf is a good resource.

Grading

- Attendance 50%
- Class performance 50%

Course policies

- You are encouraged to work in small groups, or talk to AI, but please make sure you understand the concepts yourself. A simple way to check is to write things down only relying on yourself.